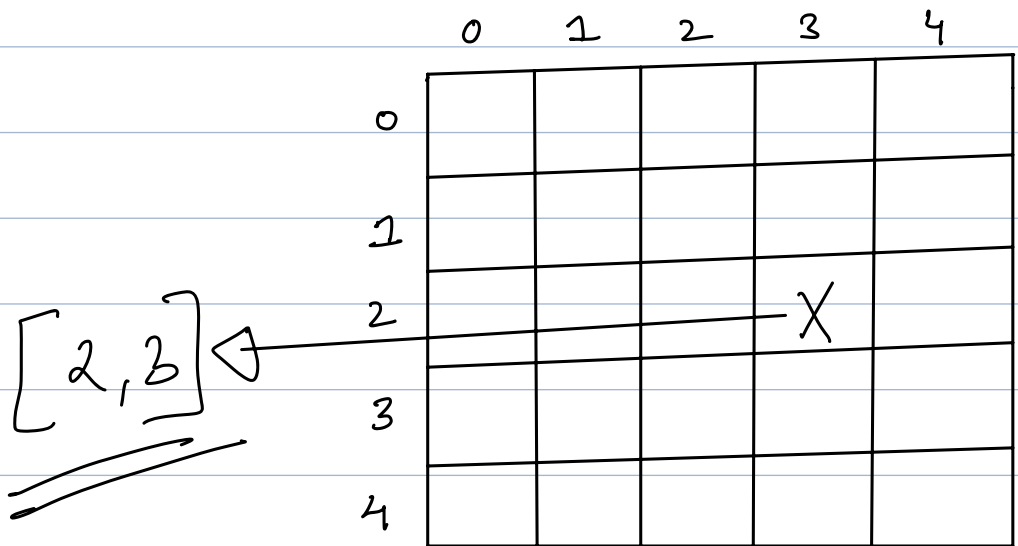
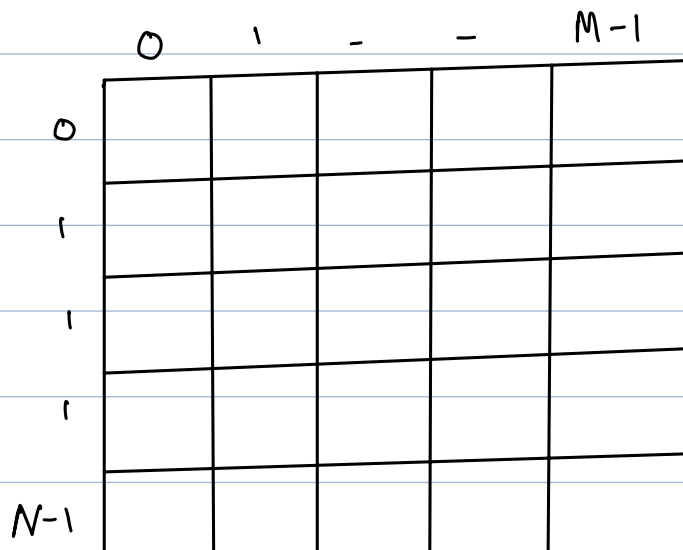


2D Arrays [BASICS]

int mat[5][5]



int mat[N][M]



Q1) Given a $\text{mat}[N][M]$, print row-wise sum.

$\text{mat}[3][4]$

	0	1	2	3	
0	3	8	9	2	= 22
1	1	2	3	6	= 12
2	4	10	11	17	= 42

```
for (int i = 0; i < n; i++) {
```

$\text{int sum} = 0;$

```
for (int j = 0; j < m; j++) {
```

$\text{sum} += \text{arr}[i][j];$

```
}
```

$\text{print}(\text{sum});$

```
}
```

Q2) Given $\text{mat}[N][M]$, find max column sum.

	0	1	2	3
0	3	8	9	2
1	1	2	3	6
2	4	10	11	17

↓ ↓ ↓ ↓
8 20 23 25

$\text{int ans} = \text{Integer.min};$

```
for (int j = 0; j < m; j++) {
```

$\text{int sum} = 0;$

```
for (int i = 0; i < n; i++) {
```

$\text{sum} += \text{arr}[i][j];$

```
}
```

$\text{ans} = \text{max}(\text{ans}, \text{sum});$

```
}
```

return ans

Q3 Given a mat $[N][N]$, print diagonals.

Case 1: Left $\rightarrow$ Right ✓

Case 2: Right $\rightarrow$ Left

mat $[4][4]$

	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12
3	13	14	15	16

= 1 6 11 16

int i = 0;

while (i < n) {

print (arr[i][i]);

i++;

}

Tc: O(N)

	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12
3	13	14	15	16

= (0, 3)

= (1, 2)

= (2, 1)

= (3, 0)

int i = 0, j = n - 1

while (i < n) {

print (arr[i][j]);

i++; j--;

}

Diagonals in a Rectangle [R-]

	0	1	2	3	4
0					
1					
2					

Q24 Given $\text{mat}[N][M]$, print all diagonals going R-L.

$\text{mat}[4][6]$

	0	1	2	3	4
0	1	2	3	4	5
1	6	7	8	9	10
2	11	12	13	14	15

$\Rightarrow$

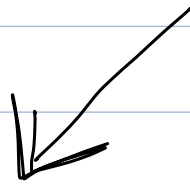
1
2 6
3 7 11
4 8 12
5 9 13
10 14
15

	0	1	2
0	1	2	3
1	4	5	6

$\Rightarrow$

1
2 4
3 5
6

	0	1	2	3	4
0	1	2	3	4	5
1	6	7	8	9	10
2	11	12	13	14	15



```
int i = 0
```

```
for (int j = 0; j < m; j++) {
```

```
    int x = i;
```

```
    int c = j;
```

```
    while (x < n && c >= 0) {
```

```
        print [arr[x][c]];
```

```
        x++; c--;
```

```
    }
```

```
}
```

```
int j = m - 1
```

```
for (int i = 1; i < n; i++) {
```

```
    int x = i;
```

```
    int c = j;
```

```
    while (x < n && c >= 0) {
```

```
        print [arr[x][c]];
```

```
        x++; c--;
```

```
    }
```

```
}
```

Q Given a 2D array, make all elements in a row or column zero if $arr[i][j] = 0$.

⇒

	0	1	2	3
0	1	2	3	4
1	5	6	7	0
2	9	2	0	4

→ (1, 3)

	0	1	2	3
0	1	2	3	4
1	5	6	7	0
2	9	2	0	4

	0	1	2	3
0	1	2	0	0
1	0	0	0	0
2	0	0	0	0

	0	1	2	3
0	1	2	3	4
1	0	2	4	0
2	9	2	0	4

	0	1	2	3
0	1	2	3	4
1	5	6	7	0
2	9	2	0	4



	0	1	2	3
0	1	2	3	4
1	5	6	7	0
2	9	2	0	4



	0	1	2	3
0	1	2	0	0
1	0	0	0	0
2	0	0	0	0

```
for (int i=0; i<n; i++) {
```

```
    int flag = 0
```

```
    for (int j=0; j<m; j++) {
```

```
        if (arr[i][j] == 0)
            flag = 1
```

```
    }
```

```
    if (flag == 1) {
```

```
        for (int j=0; j<m; j++) {
```

```
            if (arr[i][j] != 0)
```

```
                arr[i][j] = -1
```

```
        }
```

```
    }
```

```
}
```

// TODO $\Rightarrow$ Do it for columns

// TODO $\Rightarrow$ Just -1 to 0.

$T.C = O(nm)$

// taking
Case

of rows

	0	1	2	3
0				
1		0		
2				
3			0	

rows: $\{ 1, 3 \}$

cols: $\{ 1, 2 \}$